

Defect Severity Guide

For AI Defect Detection and Root-Cause Assistant using selected MVTec AD categories

Field	Details
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Decision levels	Normal, Minor, Major, Critical, Human Review Required

Important note

This document is a simulated but realistic QA severity guide created for a prototype RAG system. It is not an official MVTec document, not a certified manufacturing standard, and not a replacement for trained quality-engineering judgement. In the application, the LLM should use this guide as supporting evidence and must avoid claiming a guaranteed root cause unless supported by additional inspection data.

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1. Purpose and Scope

The purpose of this Defect Severity Guide is to define how visible anomalies and product defects should be classified by seriousness in the AI Defect Detection and Root-Cause Assistant. The guide supports the RAG module by providing clear severity rules that can be retrieved when the user asks what a detected defect means, how serious it is, and what action the operator should take.

This guide is designed around the selected MVTec AD categories used in the prototype: bottle, cable, screw, transistor, tile, toothbrush, zipper, and metal_nut. It provides practical severity decisions for common visual inspection problems such as cracks, contamination, broken parts, missing components, color or surface irregularities, thread damage, misalignment, deformation, and structural failure.

The guide should be used together with the QA Inspection Manual, Operator Action SOP, Root-Cause Troubleshooting Guide, and generated inspection report templates. It is especially useful when the AI model returns an anomaly score, predicted defect class, segmentation mask, heatmap, or confidence score, but the user needs a human-readable decision.

1.1 What this guide answers

- Is the product normal, minor defective, major defective, or critical defective?
- When should the system ask for human review instead of making a direct recommendation?
- Which defects can be reworked and which should be rejected?
- Which terms should the RAG retriever use to find severity evidence?
- How should the final inspection report phrase the severity result?

1.2 What this guide does not answer

- It does not provide official manufacturing tolerances for a real factory.
- It does not guarantee the root cause of a defect from an image alone.
- It does not replace final approval by a qualified quality engineer.
- It does not define exact dimensional thresholds unless they are visible or configured separately.

2. How Severity Is Used in the AI Inspection System

Severity is the bridge between visual detection and operational decision-making. A vision model can identify that an anomaly exists, but the factory worker also needs to know whether the product can continue, needs rework, should be rejected, or must be escalated. In this project, severity is used by four modules: the frontend result card, the LLM explanation layer, the report generator, and the agent workflow.

Module	Use of severity information
Vision model	Detects or localizes visible anomaly and returns confidence, anomaly score, heatmap, mask, or class label.
RAG retriever	Searches this guide and related SOP documents for evidence matching the predicted defect and product category.
LLM	Explains the severity in simple language using retrieved evidence and model output.
Agent	Chooses whether to recommend pass, rework, reject, retake image, or human review.
Report generator	Writes severity level, evidence, confidence, recommended action, and review status into the inspection report.

Recommended severity output format

The system should express severity in a structured way so that the UI, database, and report generator can reuse it consistently:

Field	Example value
product_category	metal_nut
defect_status	defective
severity_level	major
confidence	0.91
human_review_required	false
recommended_action	reject item and inspect machining process
evidence_source	Defect Severity Guide, Section 5.8

3. Global Severity Levels

Severity	Definition	Inspection meaning	Default action
Normal	No visible anomaly or only acceptable variation	Product may pass visual inspection	Accept
Minor	Small cosmetic issue that does not affect function, safety, sealing, assembly, or customer use	Product may be accepted with note or sent to rework depending on policy	Accept with note or rework
Major	Visible defect that may affect performance, assembly, durability, sealing, alignment, appearance quality, or customer acceptance	Product should not continue without correction	Rework if possible; otherwise reject
Critical	Defect that can cause safety risk, electrical failure, leakage, structural failure, contamination risk, sharp/broken part, or complete functional failure	Product must be removed from process immediately	Reject and escalate
Human Review Required	Image/model evidence is uncertain, conflicting, incomplete, or unsupported by documents	AI should not make final decision alone	Route to inspector

3.1 Severity principles

- Function and safety are more important than cosmetic appearance.
- Any crack, missing part, open circuit, broken thread, contamination inside a sealed product, or sharp structural damage should be treated as major or critical.
- Small surface marks may be minor only if they do not affect fit, sealing, strength, hygiene, electrical behavior, or customer acceptance.
- If severity cannot be judged from the image, the system must request human review instead of inventing a decision.
- When the model confidence is low or the image quality is poor, the severity should not be over-stated. The system should say that the result is uncertain.

4. Confidence and Human Review Rules

Severity should not be decided using visual defect type alone. The AI system must also consider model confidence, image quality, evidence availability, and whether the defect could affect safety or function. The following rules are recommended for the prototype.

Model confidence	Interpretation	Decision rule
≥ 0.90	High confidence	Allow severity decision if image quality is acceptable and RAG evidence is found.
0.75 - 0.89	Medium confidence	Allow severity suggestion but mark as recommended human verification for major or critical cases.

Model confidence	Interpretation	Decision rule
0.60 - 0.74	Low confidence	Show possible defect and request human review or clearer image.
< 0.60	Very low confidence	Do not make final severity decision. Ask for retake or manual inspection.

4.1 Human review triggers

- Confidence below 0.75 for any defect prediction.
- Image is blurry, dark, overexposed, cropped, strongly shadowed, or product is not fully visible.
- Anomaly is close to safety-critical area, sealing edge, thread, electrical connector, or functional interface.
- RAG retriever cannot find a matching severity rule in the documents.
- The system predicts critical severity for any category.
- The same image produces inconsistent results across repeated analysis.
- OCR text, batch code, or label is required but unreadable.

5. Category-Specific Severity Guidance

The following sections define severity for each selected category. These rules are written in a RAG-friendly style so that retrieval can match user questions such as 'how serious is a cracked bottle?', 'what action for broken zipper teeth?', or 'is a scratched metal nut acceptable?'

5.1 Bottle

Bottle inspection focuses on structural integrity, contamination risk, sealing ability, and customer-visible quality. Any damage that can cause leakage, sealing failure, contamination, or breakage is treated seriously.

Observed condition	Severity	Recommended action	Reason
Small cosmetic mark outside sealing area	Minor	Accept with note or rework if appearance standard requires it	No leakage risk and no crack visible.
Surface scratch on body	Minor to Major	Rework or reject depending on depth and visibility	Major if deep, near sealing area, or customer-facing surface.
Crack on body, base, neck, or edge	Critical	Reject and escalate	Crack can cause leakage, breakage, or contamination.
Broken/chipped rim or sealing lip	Critical	Reject and inspect molding/handling process	Sealing quality can fail.
Foreign particle, stain, or contamination	Major to Critical	Hold product; clean/reject based on contamination type	Critical if inside bottle or hygiene-sensitive.
Deformation or abnormal shape	Major	Reject or route to supervisor	May affect filling, capping, handling, or sealing.
Missing section or severe break	Critical	Reject immediately	Structural failure.

RAG keywords: bottle, crack, broken, contamination, leakage, sealing, rim, neck, deformation, chip, foreign particle

5.2 Cable

Cable inspection focuses on insulation integrity, conductor exposure, connector condition, routing shape, and contamination. Electrical safety risks should be treated as critical.

Observed condition	Severity	Recommended action	Reason
Small surface dirt or light mark	Minor	Clean or accept with note	Only if insulation is intact.
Outer insulation scratch	Minor to Major	Rework or reject based on depth	Major if conductor protection may be weakened.
Cut, tear, or exposed conductor	Critical	Reject and escalate	Electrical safety failure.
Connector damage or missing pin	Critical	Reject	Functional electrical connection can fail.
Cable misrouting, bend, or deformation	Major	Hold for rework or reject	May cause strain or assembly failure.

Observed condition	Severity	Recommended action	Reason
Contamination on connector/contact area	Major to Critical	Clean/hold; reject if severe	May cause contact resistance or electrical failure.
Missing section or broken wire	Critical	Reject immediately	Complete functional failure risk.

RAG keywords: cable, insulation, exposed conductor, connector, missing pin, contamination, wire, electrical, bend, cut

5.3 Screw

Screw inspection focuses on thread quality, head geometry, shaft straightness, corrosion, and damage that may affect fastening. Thread and head defects can prevent proper assembly.

Observed condition	Severity	Recommended action	Reason
Tiny cosmetic discoloration	Minor	Accept with note	Only if no corrosion or thread damage.
Light surface scratch	Minor	Accept or rework	Does not affect thread or head engagement.
Damaged thread	Major	Reject or rework if re-threading is allowed	May prevent correct fastening.
Broken thread or missing thread section	Critical	Reject	Fastening failure risk.
Damaged screw head/drive slot	Major to Critical	Reject if tool engagement is affected	Critical when drive cannot engage.
Bent shaft or deformation	Critical	Reject immediately	Assembly and load-bearing failure risk.
Corrosion/rust	Major	Hold and evaluate; reject if severe	Can reduce strength and customer acceptance.

RAG keywords: screw, thread, head, slot, bent, corrosion, rust, damaged thread, fastening, shaft

5.4 Transistor

Transistor inspection focuses on component body integrity, pin/lead condition, alignment, contamination, and assembly correctness. Electrical components require strict severity handling because defects may cause circuit failure.

Observed condition	Severity	Recommended action	Reason
Small surface mark on package	Minor	Accept with note if markings and body are intact	No electrical or assembly impact visible.
Scratched or unreadable marking	Major	Hold for manual verification	Traceability may be affected.
Bent lead/pin	Major	Rework if alignment can be corrected; otherwise reject	May affect soldering or insertion.
Broken or missing lead/pin	Critical	Reject	Electrical connection failure.
Cracked package/body	Critical	Reject and escalate	Internal component damage possible.
Contamination on leads/contact area	Major to Critical	Clean/hold; reject if severe	May affect conductivity or solderability.
Wrong orientation or component mismatch	Critical	Reject or rework under supervisor	Circuit function and safety risk.

RAG keywords: transistor, lead, pin, package, marking, body crack, bent lead, missing lead, solder, contact

5.5 Tile

Tile inspection focuses on cracks, chips, surface contamination, pattern defects, glaze damage, and dimensional appearance. Some tile defects are cosmetic, but cracks and broken edges are serious.

Observed condition	Severity	Recommended action	Reason
Small color variation within allowed shade	Minor	Accept with note	Only if within appearance tolerance.
Small surface stain or dust	Minor	Clean or rework	If removable and not embedded.
Visible scratch or glaze damage	Minor to Major	Rework or reject depending on visibility	Major for deep or customer-facing damage.
Crack across surface or edge	Critical	Reject	Structural and customer-quality failure.

Observed condition	Severity	Recommended action	Reason
Broken corner or chipped edge	Major to Critical	Reject if visible or sharp	Critical if sharp, deep, or affects laying.
Pattern/texture anomaly	Major	Reject or route to grading	May fail appearance standard.
Warping or deformation	Major	Reject	May affect installation quality.

RAG keywords: tile, crack, chip, glaze, scratch, color, stain, pattern, texture, broken corner, edge

5.6 Toothbrush

Toothbrush inspection focuses on hygiene, bristle condition, handle integrity, deformation, missing parts, and sharp or unsafe edges. Because toothbrushes contact the mouth, contamination and structural defects must be handled strictly.

Observed condition	Severity	Recommended action	Reason
Minor handle color variation	Minor	Accept with note	Only if no hygiene or functional issue.
Light cosmetic mark on handle	Minor	Clean/rework or accept	No sharp edge or contamination.
Damaged or missing bristles	Major	Reject or rework if allowed	Cleaning performance affected.
Foreign particle or contamination	Critical	Reject and escalate hygiene check	Mouth-contact safety risk.
Cracked handle or sharp edge	Critical	Reject	User injury risk.
Deformed head or bristle field	Major	Reject	Function and appearance affected.
Loose head/handle separation	Critical	Reject immediately	Structural safety risk.

RAG keywords: toothbrush, bristle, handle, contamination, hygiene, sharp edge, cracked handle, missing bristles, deformed head

5.7 Zipper

Zipper inspection focuses on tooth integrity, slider function, tape alignment, missing teeth, broken parts, contamination, and smooth closing behavior. Defects that prevent closing/opening are serious.

Observed condition	Severity	Recommended action	Reason
Small cosmetic mark on tape	Minor	Accept or clean	No functional issue.
Misaligned tape or tooth row	Major	Rework or reject	May affect closing quality.
Broken or missing zipper tooth	Critical	Reject	Zipper may fail to close.
Damaged slider/pull tab	Major to Critical	Reject if function affected	Critical if zipper cannot operate.
Stuck zipper or irregular sliding	Major	Rework or reject	Functional quality issue.
Torn fabric tape	Critical	Reject	Structural failure.
Contamination in tooth path	Major	Clean/hold; reject if embedded	May block operation.

RAG keywords: zipper, tooth, teeth, slider, pull tab, tape, misalignment, stuck, broken tooth, missing tooth

5.8 Metal Nut

Metal nut inspection focuses on thread integrity, bore quality, surface defects, corrosion, deformation, and ability to fasten safely. Thread and shape issues can directly affect assembly and load-bearing behavior.

Observed condition	Severity	Recommended action	Reason
Light surface discoloration	Minor	Accept with note if no corrosion	Cosmetic only.
Small surface scratch outside thread	Minor	Accept or rework	No effect on fastening.
Internal thread damage	Major	Reject or rework if thread repair is approved	May prevent correct assembly.
Missing/deformed thread section	Critical	Reject	Fastening failure risk.
Crack on nut body	Critical	Reject and escalate	Structural failure risk.
Corrosion/rust	Major	Hold and evaluate; reject if severe	Can reduce strength and customer acceptance.

Observed condition	Severity	Recommended action	Reason
Deformed shape or bore	Major to Critical	Reject if fit is affected	May not fit bolt or tool.
Foreign material inside bore	Major	Clean/hold; reject if embedded	Can block assembly.

RAG keywords: metal nut, thread, internal thread, bore, corrosion, rust, crack, deformation, fastening, foreign material

6. Cross-Category Defect Severity Matrix

The table below provides a quick rule-of-thumb severity mapping that can be retrieved by RAG when the exact product-specific section is not found. Category-specific rules should be preferred when available.

Defect family	Default severity	Escalation rule
Crack	Major to Critical	Critical if on load-bearing, sealing, electrical, hygiene, or structural area.
Scratch	Minor to Major	Minor if cosmetic only. Major if deep, sharp, on sealing/contact area, or customer-critical surface.
Contamination	Minor to Critical	Minor if removable surface dust. Critical for hygiene products, electrical contacts, sealed containers, or internal contamination.
Missing part	Critical	Reject when a required structural, electrical, sealing, or functional part is missing.
Deformation	Major to Critical	Critical if it prevents fit, function, sealing, fastening, or safe use.
Color/appearance anomaly	Minor to Major	Major when customer-visible, indicates material issue, or violates grade standard.
Thread damage	Major to Critical	Critical when fastening engagement is impossible or unsafe.
Broken edge/chip	Major to Critical	Critical if sharp, sealing area, structural area, or hygiene-sensitive.
Electrical contact defect	Critical	Reject because conductivity, safety, or circuit function may fail.
Blurry image/uncertain evidence	Human Review Required	Retake image or send to inspector. Do not overstate severity.

7. Root-Cause Severity Hints

This guide supports possible root-cause suggestions, but the system must not present root cause as guaranteed from image evidence alone. Root cause should be phrased as 'possible cause', 'likely inspection checkpoint', or 'recommended process check'.

Observed defect	Possible causes	Recommended process check
Bottle cracks or chips	High molding stress, poor cooling, impact during handling, damaged conveyor guide	Check molding parameters, cooling stage, and handling impact points
Cable insulation cut	Blade/contact damage, poor routing, excessive bending, handling damage	Inspect cutting tools, routing path, clamp pressure, and storage handling
Screw thread damage	Worn die/tool, misalignment, incorrect feed rate, material defect	Inspect threading tool, alignment, and sample more parts from same batch
Transistor bent/missing leads	Feeder jam, pick-and-place misalignment, poor packaging, handling impact	Check feeder alignment, placement head, tray handling, and lead protection
Tile cracks/chips	Impact, thermal stress, poor firing/cooling, packaging damage	Check kiln/cooling profile, transport vibration, and packaging protection
Toothbrush bristle defects	Bristle insertion problem, trimming issue, contamination during packaging	Inspect bristle insertion machine and hygiene control points
Zipper tooth/slider damage	Misalignment, worn mold/tool, poor tape tension, assembly pressure issue	Check tooth forming process, slider assembly station, and tape tension
Metal nut thread/corrosion	Machining error, worn tap, poor coating, moisture exposure	Inspect tapping tool, coating process, humidity control, and batch storage

8. Operator Decision Rules

The following action logic should be used by the AI agent and final report. The agent should combine model confidence, severity, category, and RAG evidence before selecting an action.

Severity	Default operator action	When to use
Normal	Accept product	No anomaly detected and image quality acceptable.
Minor	Accept with note or send to rework	Use when defect is cosmetic and does not affect function or safety.

Severity	Default operator action	When to use
Major	Hold item; rework if approved; otherwise reject	Use when defect may affect performance, assembly, appearance standard, or customer acceptance.
Critical	Reject item and escalate to supervisor	Use when defect affects safety, sealing, electrical contact, hygiene, structure, or core function.
Human Review Required	Send to inspector or request new image	Use when confidence/evidence is weak, image is poor, or decision is high risk.

8.1 Action phrases for LLM output

- Normal: 'No significant defect is visible. The item can proceed if other quality checks pass.'
- Minor: 'A minor visual issue is present. The item may be accepted with note or routed for cosmetic rework depending on the product grade.'
- Major: 'The defect may affect quality or function. Hold the item and apply the relevant rework or rejection procedure.'
- Critical: 'The defect can affect safety, sealing, electrical function, hygiene, or structural integrity. Reject the item and escalate to a supervisor.'
- Human Review Required: 'The system cannot make a reliable decision from the current evidence. A human inspector should review the item or a clearer image should be provided.'

9. RAG Retrieval Keywords and Expected Evidence

For RAG, each document chunk should include category names, severity terms, defect synonyms, and action terms. This improves retrieval when the user asks natural-language questions.

Category	Useful retrieval keywords	Expected evidence/result
Bottle	bottle crack, rim chip, sealing lip, leakage, contamination, deformation	Reject cracked or chipped sealing surfaces; inspect molding and handling.
Cable	insulation cut, exposed conductor, connector pin, electrical contact, contamination	Reject exposed conductors and connector damage; inspect electrical safety.
Screw	thread damage, screw head, bent shaft, rust, corrosion, fastening	Reject broken thread, bent screw, unusable head, or severe corrosion.
Transistor	bent lead, missing pin, body crack, unreadable marking, contact contamination	Reject missing/broken leads or cracked package; verify traceability.
Tile	tile crack, chipped edge, glaze scratch, color variation, pattern anomaly	Reject cracks and sharp chips; grade cosmetic appearance issues.
Toothbrush	bristle damage, contamination, cracked handle, sharp edge, hygiene	Reject contamination, sharp edges, and broken handle/head.
Zipper	missing tooth, broken tooth, slider damage, tape tear, stuck zipper	Reject missing teeth, torn tape, and nonfunctional slider.
Metal nut	internal thread, bore, nut crack, corrosion, deformation, fastening	Reject thread failure, cracks, deformation affecting fit, and severe corrosion.

10. Example Severity Assessments

Scenario	Correct severity decision
Input: bottle image with visible crack near neck. Model confidence: 0.94.	Severity: Critical. Action: Reject and escalate. Evidence: cracks near neck/sealing area can cause leakage and structural failure.
Input: cable image with small surface dirt but insulation intact. Confidence: 0.88.	Severity: Minor. Action: Clean or accept with note. Evidence: cosmetic dirt only if insulation and connector are intact.
Input: screw image with missing thread section. Confidence: 0.91.	Severity: Critical. Action: Reject. Evidence: missing thread can cause fastening failure.
Input: transistor image with bent lead. Confidence: 0.82.	Severity: Major. Action: Rework if alignment can be corrected; otherwise reject. Human verification recommended.

Scenario	Correct severity decision
Input: tile image with small dust mark. Confidence: 0.86.	Severity: Minor if removable. Action: Clean/rework. If embedded stain, escalate to major.
Input: toothbrush with foreign particle near bristles. Confidence: 0.89.	Severity: Critical. Action: Reject and escalate hygiene check.
Input: zipper with missing tooth. Confidence: 0.95.	Severity: Critical. Action: Reject. Evidence: closure function can fail.
Input: metal nut with light surface scratch outside thread. Confidence: 0.90.	Severity: Minor. Action: Accept with note or cosmetic rework if appearance standard requires.
Input: blurry image; possible anomaly but confidence 0.58.	Severity: Human Review Required. Action: request clearer image or route to inspector.

11. Report Writing Rules

The generated inspection report should be clear, traceable, and conservative. It should not exaggerate certainty. It should connect the visual result with retrieved evidence and give a practical action.

Report field	Writing rule
Product category	Use selected MVTec category name, for example 'bottle' or 'metal_nut'.
Defect status	Use 'normal', 'defective', or 'uncertain'.
Severity level	Use exactly one of: Normal, Minor, Major, Critical, Human Review Required.
Evidence	Mention the retrieved guide section or manual statement in summary form.
Recommended action	Accept, clean, rework, hold, reject, escalate, or request clearer image.
Root cause	Use 'possible root cause' wording, not guaranteed cause.
Human review	State yes/no and explain why.

11.1 Good report wording

'The uploaded metal_nut image shows a likely internal thread anomaly with high model confidence. According to the severity guide, thread damage is a major defect and missing or deformed thread sections can be critical because fastening may fail. Recommended action: hold the item for inspection and reject if thread engagement is affected. Possible process checkpoint: tapping tool wear or alignment issue.'

11.2 Wording to avoid

- Avoid saying: 'The cause is definitely machine failure.' Use: 'Possible cause may be machine/tool wear.'
- Avoid saying: 'This product is safe' when only image inspection was done. Use: 'No visible defect was detected in this image.'
- Avoid final rejection when confidence is low. Use: 'Human review required.'
- Avoid citing a document section that was not actually retrieved.

12. Limitations and Safety Notes

- Image-only inspection cannot measure internal defects, invisible material weakness, electrical performance, exact dimensions, chemical contamination, or fatigue strength.
- The system should treat severe functional, safety, hygiene, sealing, and electrical issues conservatively.
- If the detected severity is critical, the output should request supervisor or human QA review even when model confidence is high.
- If the RAG system retrieves no relevant evidence, the LLM should say that supporting evidence was not found and recommend human review.
- For production use, this guide must be replaced or validated by real company QA standards, engineering drawings, tolerances, and customer requirements.

13. RAG Chunking Guidance

When loading this guide into a vector database, chunk by section and keep headings with the text. Category-specific tables should be chunked so each category keeps its defect terms, severity, action, and reason in the same chunk.

RAG setting	Recommendation
Chunk size	500 to 900 tokens per chunk for detailed retrieval.
Overlap	80 to 150 tokens so category heading and severity terms are preserved.
Metadata	document_name, section_title, category, severity_level, defect_family.
Embedding content	Include headings, table rows, and keywords. Do not embed only the table cell without context.
Retriever query expansion	Add synonyms such as crack/break/chip, contamination/foreign particle/dirt, thread damage/fastening failure.

Suggested metadata examples

Metadata key	Example
document_name	Defect_Severity_Guide.pdf
section_title	5.8 Metal Nut
category	metal_nut
defect_family	thread damage
severity_level	major/critical
recommended_action	reject or rework only if approved

14. Sample Questions for Testing Retrieval

Use these questions to test whether the RAG retriever can find the correct section before the LLM writes the answer.

- 1 How serious is a crack near the bottle neck?
- 2 What should we do if a cable has exposed conductor?
- 3 Is a small cosmetic mark on a screw critical?
- 4 What severity should be assigned to a transistor with a missing lead?
- 5 Can a cracked tile be accepted?
- 6 What action is needed for toothbrush bristle contamination?
- 7 Is a zipper with missing teeth reject-level?
- 8 What does the guide say about damaged internal threads in a metal nut?
- 9 When should human review be required?
- 10 How should the report describe possible root cause?

End of document

This Defect Severity Guide is intended to be used as a RAG document together with the QA Inspection Manual, Operator Action SOP, Root-Cause Troubleshooting Guide, Inspection Report Template, Image Quality Check SOP, and Human Review and Escalation Policy.